Software Architecture

Scenario Analysis and Classroom Exercise

Q1. Online Movie Streaming

A movie streaming platform allows users to watch movies using their phones, laptops, and smart TVs. All devices request movie information and streaming services from remote servers.

Which architectural pattern best describes this system? Why?

Expected answer: Client–Server Architecture

Phones, laptops, and smart TVs act as clients. They send requests to remote servers, which provide movie information and streaming services.

Q2. Online Shopping Discount

An online shopping application separates its user interface, discount calculations, and database access. The company frequently changes its discount rules but wants to avoid modifying the user interface or database access code.

Which architectural pattern would you recommend? Why?

Expected answer: Layered Architecture

The presentation layer shows the shopping interface; the business logic layer applies discount rules; the data access layer reads or saves data. Changing the discount rules can be localized to the business logic layer.

Q3. Student Grade System

A university application displays the same student grades in different ways: students see a grade report, while teachers see charts. When grades change, both displays should reflect the updated information.

Which architectural pattern is most suitable? Identify the responsibilities of its main components.

Expected answer: Model–View–Controller (MVC)

Model: manages grades and the operations on them. View: presents grades as a report or chart. Controller: handles actions such as selecting a student or requesting a view. Multiple views can use the same grade data.

Q4. Software Development Tools

A software company uses several tools: a UML editor, a Java code editor, a code generator, and a report generator. All tools need to access and update the same project information without communicating directly with one another.

Which architectural pattern would be most suitable? Explain how these tools share information.

Expected answer: Repository Architecture

All tools read project information from, and write updates to, a shared project repository. They do not have to exchange information directly with one another.

Q5. Employee Salary Processing

A company develops a salary processing application. Every month, employee records pass through several steps: checking attendance, calculating salaries, deducting taxes, and generating payslips. Each step processes the output from the previous step.

Which architectural pattern is most appropriate? Why.

Expected answer: Pipe-and-Filter Architecture

Each processing stage performs one transformation, then passes the result to the next stage: attendance checking → salary calculation → tax deduction → payslip generation.

Q6. University Course Registration System

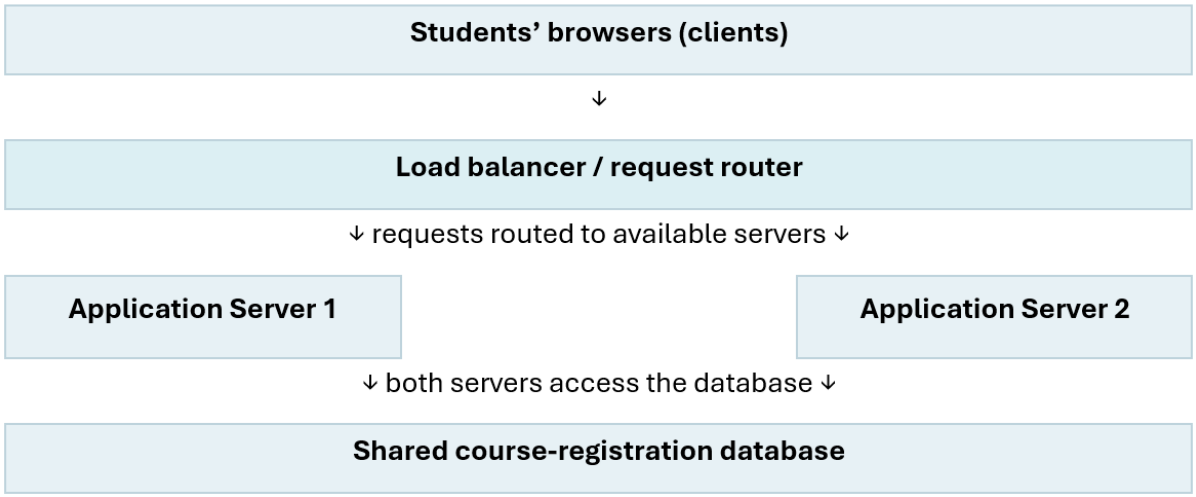
A university is developing an online course registration system. Students access the system through their browsers to register for courses.

During registration week, thousands of students may access the system simultaneously. The university wants the system to continue working even if one application server fails.

Tasks:

- Draw a simple high-level architecture showing the students' browsers, application servers, and database.
- Suggest one architectural decision to improve the system's availability.
- Explain your decision in one or two sentences.

Expected Answer:



- Architecture: Students use web browsers (clients). Requests go through a load balancer to an application server; the server reads or updates a shared database.
- Availability decision: Use at least two application servers and direct requests only to a server that is working.
- Justification: If one application server fails, the load balancer can send requests to the other server, helping the system remain accessible.